PILOT PROTOCOL · DRAFT

Recording a knot as it lets go: a pilot protocol

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Aim

To record, for the first time, what happens in the tissue at the moment a knot lets go, and so tell apart the seven mechanisms proposed for knots. Each has been run as a model from its own physiology; at a release each would leave a different trace. This pilot tests feasibility and the direction of those traces, not their size.

What each theory predicts

Theory	Recorded at release	Under a resting hand	Where knots sit
結 Perforators	skin flow floods in	as the hand lifts (never under it)	at the fascia, on its vessels
閂 Vascular latch	flow rises slowly	under the hand	anywhere there are small arteries
点 Trigger points	stiffness falls in the muscle	under the hand	in muscle, where its nerve enters
膠 Densification	the layers slide again	not within a minute (never under it)	in the gliding layer; broad
神経 Nerves	the nerve's firing falls	not within a minute	fixed points, at nerves
握 Motor switch	a motor unit falls silent	under the hand	in muscle, anywhere
覚 Perception	nothing in the tissue	not within a minute (never under it)	anywhere; broad

From the models' runs in 32 settings each; a perforator's sleeve adds that the layers slide again only once moved, not at the release itself.

Participants and setting

Twelve adults who report knots in the upper back and notice their releases. Excluded: cardiovascular or peripheral vascular disease, neuropathy, pregnancy, current psychiatric crisis, and any history of severe difficulty with meditation or bodywork. Ethics approval and informed consent; a quiet laboratory at 24 °C after 20 minutes' acclimatisation, seated.

Instruments

Laser speckle contrast imaging of skin flow over a 10 × 10 cm field of the upper back; high-density surface EMG (a 64-electrode grid) over the upper trapezius, decomposed into single motor units; ultrasound (B-mode, colour Doppler and shear-strain imaging of the layers' sliding); a pressure sensor under the practitioner's hand; and a button the participant presses at each felt release.

Procedure (about 90 minutes)

1. **Map.** The participant marks 3–5 knots; a blinded examiner maps perforators by colour Doppler; ultrasound notes the layer of any bump; two control points without knots are marked.
2. **Baseline.** Five minutes at rest, all instruments recording.
3. **Release trials,** each knot in random order, at the participant's usual pace: a practitioner's hand resting at comfortable pressure for up to a minute, then lifted; attention without touch for up to three minutes; a roller over the knot for up to three minutes. The same hand and roller on a control point (the pressed sham). Three minutes' rest between trials.
4. **After each release:** ultrasound of the layers' sliding at 0, 5 and 10 minutes, and after the participant moves the shoulder.

Outcomes

Locked to each button press (30 s either side): skin flow at the knot against the pressed sham; any motor unit falling silent under the grid; the layers' shear strain at the knot; whether knots sit on perforators more often than control points; and how long after a release the layers slide again. Twelve people with 3–5 knots give 40–60 release events, enough to see the direction of each trace.

Reading the result

Skin flow flooding in at the knot and not at the sham: perforators. Flow rising slowly over tens of seconds: the vascular latch. A motor unit falling silent at the felt release: the motor switch. Stiffness falling in the muscle under the press: trigger points. The layers sliding again at the release: densification; only after the shoulder moves: a perforator's sleeve. Nothing in the tissue: perception, or a nerve, which a local anaesthetic at the point would then test (in a clinical setting). Different knots may give different traces: the pilot is designed to find that too.

Safety

Ordinary breathing throughout: no breath-holds or forceful breathing. Comfortable pressure only, measured, and nothing on the front of the neck. Any trial stops at the first sign of distress. Deep somatic and contemplative work can cause lasting difficulties; see the site's moderation guidance. The predictions and analysis are registered before any recording.